PRECURSA

Self-Healing Supply Chain Intelligence

Complete Project Documentation & Blueprint

Team Return0 | Smart Supply Chains Track

April 2026

$184B	< 2 min	85%	$23.6B
Annual Losses	Detection Time	Detectable Signals	Market by 2028

1. Executive Summary

Precursa is an AI-powered self-healing supply chain intelligence platform that detects, predicts, and autonomously resolves disruptions before they cascade into network-wide failures. In a world where $184 billion is lost annually to supply chain disruptions — and 73% of companies still rely on manual monitoring — Precursa delivers predictive intelligence with a detection time under 2 minutes, compared to the industry average of 4–6 hours.

The platform ingests 10+ live data streams (weather APIs, port feeds, IoT sensors, carrier GPS, geopolitical risk indices, and customs APIs), runs a 3-model AI ensemble to generate a per-shipment Disruption Risk Index (DRI), and uses a graph-based rerouting engine to autonomously select optimal alternate routes — all without human intervention.

Detect	Predict	Heal
Sub-2-minute disruption detection via continuous AI monitoring	Hours-ahead forecasting using LSTM temporal models on live data	Autonomous rerouting, carrier rebooking & customer notification

Built for the Smart Supply Chains hackathon track, the 50-hour executable MVP demonstrates the full detect-predict-heal loop through a live dashboard, real-time DRI scoring, interactive rerouting simulation, and an AI Copilot powered by the Claude API.

2. Problem Statement

2.1 The Core Problem

Modern supply chains are fundamentally reactive. Disruptions — whether caused by extreme weather, port congestion, geopolitical events, or customs delays — are typically detected only after timelines have already been missed. By the time a logistics manager is alerted, the disruption has already cascaded into delayed inventory, missed SLAs, and mounting losses.

2.2 Scale of the Problem

Metric	Current Reality	Impact	Root Cause
Detection Time	4–6 hours average	Cascading delays	Manual monitoring
Monitoring Method	73% manual	$184B annual loss	No unified data layer
Disruption Visibility	Localized only	Network-wide failure	Siloed systems
Recovery	Manual rerouting	2–3 day delays	No automation layer

2.3 Three Core Pain Points

Pain Point 1: Reactive, Not Predictive

Critical disruptions — weather events, port delays, carrier bottlenecks — are detected only AFTER timelines are already missed. No current system correlates early-warning signals across data streams to provide advance notice.

Pain Point 2: Unmanageable Scale

Global networks process millions of concurrent shipments. Manual teams can only monitor a fraction. No existing tool connects weather data, IoT sensors, carrier GPS, geopolitical risk, and customs status into a unified real-time intelligence layer.

Pain Point 3: Cascading Failures

A single localized bottleneck — one congested port, one delayed carrier — silently snowballs into network-wide delays worth millions in losses. Without automated propagation analysis, companies cannot see the cascade coming.

2.4 The Opportunity

85% of supply chain disruptions have detectable precursor signals in data that already exists — weather APIs, AIS vessel tracking, port authority feeds, news sentiment APIs. The problem is not data availability; it is the absence of an intelligent layer that aggregates, correlates, and acts on this data in real time.

3. Solution Overview

3.1 What Precursa Does

Precursa is the intelligence layer that sits between raw supply chain data and autonomous action. It ingests 10+ live data streams, runs a three-model AI ensemble to score disruption risk per shipment, uses a graph-based rerouting engine to compute optimal alternate routes, and executes autonomous remediation — rebooking carriers, updating warehouse slots, and notifying customers — before the disruption reaches critical impact.

3.2 Four Core Pillars

#	Pillar	What It Does	Key Technology
01	Omni-Source Data Fabric	Ingests weather, port feeds, IoT sensors, carrier GPS, geopolitical risk, and customs APIs into a unified stream	Apache Kafka, Confluent Cloud
02	AI Disruption Detection	Three-model ensemble generates a Disruption Risk Index (DRI) per shipment, updated every 2 minutes	PyTorch LSTM + XGBoost + Isolation Forest
03	Graph-Based Rerouting Engine	Live supply chain graph computes optimal alternate routes ranked by cost, ETA, risk, and carbon footprint	Neo4j + NetworkX + Modified Dijkstra
04	Self-Healing Automation	Autonomously books alternate carriers, updates warehouse slots, and notifies customers with revised ETAs	LangGraph + Claude API by Anthropic

3.3 The Disruption Risk Index (DRI)

The DRI is Precursa's core output — a 0–100 risk score per shipment, updated every 2 minutes. It is the ensemble product of three model outputs:

- LSTM temporal score: Probability of disruption in the next 6 hours based on historical sequences
- Isolation Forest anomaly score: Deviation from normal shipment behavior patterns
- XGBoost classification score: Risk category based on current contextual features

DRI 0–30	Green — Normal operations, no action required
DRI 31–60	Yellow — Monitor closely, prepare contingency routes
DRI 61–80	Orange — High risk, recommend operator review

DRI 81–100	Red — Critical, autonomous rerouting triggered

3.4 AI Copilot

Every disruption event and rerouting decision is explained in plain language by an AI Copilot powered by the Claude API. Operators can ask questions like 'Why was this route chosen?' or 'What is the cost-carbon tradeoff on SHP-004?' and receive a contextual, data-grounded answer in real time. This transforms Precursa from a black box into a transparent, collaborative intelligence partner.

4. Technical Architecture

4.1 Full Vision Architecture (18-Month)

The complete Precursa architecture as designed for production deployment:

Data Streaming	Apache Kafka + Confluent Cloud — real-time pub/sub for 10+ live data streams
Stream Processing	Apache Flink — sub-second stateful stream processing with event windowing
ML Models	PyTorch LSTM + XGBoost + Isolation Forest — 3-model ensemble with MLflow tracking
ML Ops	AWS SageMaker — model monitoring, continuous retraining, feedback loop
Graph Engine	Neo4j + NetworkX — live supply chain graph with modified Dijkstra algorithm
LLM Copilot	Claude API by Anthropic — natural language explanation and Q&A interface
Automation Agent	LangGraph — agentic orchestration for autonomous rerouting actions
Backend / API	FastAPI + WebSockets — real-time data push to frontend
Frontend	React + Mapbox GL JS — real-time shipment map, DRI heatmap, alerts
Infrastructure	Docker + Kubernetes (EKS) + Terraform — cloud-native deployment

4.2 50-Hour MVP Architecture (Hackathon Build)

The following is the executable architecture scoped for the hackathon demo. Every full-vision component has a pragmatic replacement that delivers the same demo experience without the setup overhead:

Component	Full Vision	50-Hour MVP Replacement	Why
Event Bus	Apache Kafka + Confluent Cloud	FastAPI mock data generator	Kafka setup alone takes 6–8 hrs
Stream Processing	Apache Flink	Python background polling loop (2-min interval)	Same effect, zero infra
ML Ensemble	LSTM + Isolation Forest + XGBoost	Single XGBoost model on Kaggle logistics dataset	Trains in minutes, still impressive

Component	Full Vision	50-Hour MVP Replacement	Why
Graph Engine	Neo4j + NetworkX	NetworkX only (pure Python)	Neo4j setup is unnecessary for demo
Automation Agent	LangGraph orchestration	Direct Claude API structured prompt	Same autonomous feel, fraction of work
ML Ops	AWS SageMaker	Local model served via FastAPI endpoint	No cloud setup needed for demo
Frontend Maps	React + Mapbox GL JS	React + Leaflet.js (free, no API key friction)	Mapbox billing setup wastes time
Infrastructure	Docker + Kubernetes + Terraform	Run locally on laptop	Nobody checks infra during demo

4.3 System Flow (MVP)

The following describes the end-to-end data and control flow in the MVP:

1. Mock data generator produces 5 shipments with real-world port coordinates, cargo type, and DRI scores every 30 seconds
2. FastAPI backend receives data, runs XGBoost model to update DRI per shipment
3. WebSocket pushes updated shipment state to React frontend in real time
4. Frontend renders shipment markers on Leaflet map with color-coded DRI badges
5. Operator clicks 'Simulate Disruption' — backend triggers DRI spike for selected shipment
6. NetworkX graph engine computes 3 alternate routes ranked by cost, ETA, risk, and carbon score
7. Auto-rerouting fires: route updates on map, carrier reallocation logged
8. Claude API Copilot generates natural language explanation of the decision
9. Customer notification panel shows revised ETA and reroute rationale

4.4 API Endpoint Design

GET /api/shipments	Returns all active shipments with current DRI scores, coordinates, and status
POST /api/trigger-disruption	Simulates a disruption event on a specified shipment (type: port/weather/customs)
POST /api/reroute/{id}	Triggers NetworkX graph rerouting, returns top 3 alternate routes
GET /api/routes/{id}	Returns route graph for a shipment with node metadata

POST /api/copilot	Sends disruption context to Claude API, returns natural language explanation
WS /ws/live	WebSocket endpoint streaming real-time shipment state updates

5. Demo Blueprint

The demo is the single most important element of the hackathon presentation. This section defines exactly what judges will see, in what order, and what talking points to deliver at each moment. Target runtime: 2.5 to 3 minutes.

5.1 Pre-Demo Setup Checklist

- Backend running on localhost:8000 with 5 seeded shipments
- Frontend running on localhost:3000 — map loaded, all shipments visible
- Claude API key loaded in environment, Copilot panel visible
- Browser zoomed to 90% — full dashboard visible without scrolling
- Disable all browser notifications and OS alerts
- Have the 3 disruption scenarios pre-selected in the simulate panel

5.2 Demo Script — Step by Step

Step	Action	What Judges See	Talking Point
1	Open Dashboard	Global map with 5 live shipment markers. Green/yellow DRI score badges on each.	'This is Precursa's Ops Dashboard. 5 shipments active across 3 trade lanes. All green right now — DRI scores below 30.'
2	Trigger Disruption	Click 'Simulate: Port Congestion — Singapore'. SHP-004's DRI spikes from 23 to 87. Marker turns red.	'We just simulated a port congestion signal hitting Singapore. Watch SHP-004's DRI spike in real time — that's the AI ensemble firing.'
3	Alert Fires	Toast notification: '⚠ Disruption detected on SHP-004. Risk Index: 87. Cause: Port congestion.'	'Under 2 minutes from signal to alert. Industry average is 4 to 6 hours. We've already caught it.'
4	Rerouting Triggers	Route on map visibly shifts. Sidebar shows 3 alternate routes with cost, ETA, risk, and carbon scores.	'Graph engine is running modified Dijkstra across 18 port nodes. Three routes ranked by our composite score. Best route auto-selected.'
5	Copilot Explains	Claude Copilot panel: 'Rerouted SHP-004 via Colombo. ETA impact: +6 hours. Cost delta: -$1,200. Carbon reduced 8%.'	'Every autonomous decision is explained. Operators stay in control — they can override or ask follow-up questions.'
6	Ask Copilot Live	Type 'Why was Colombo chosen over Jebel Ali?' — Copilot explains cost and congestion tradeoff.	'Watch this.' [Ask question live.] This is the moment that makes Precursa feel like a real product, not a prototype.'

5.3 Three Demo Scenarios (Prepared)

Have three disruption scenarios seeded and ready to trigger during the demo:

- Scenario A — Port Congestion (Singapore): DRI spike on Mumbai-Rotterdam lane. Best reroute via Colombo.
- Scenario B — Weather Event (Bay of Bengal): DRI spike on Chennai-Hamburg lane. Best reroute via Colombo overland segment.
- Scenario C — Customs Delay (Suez): DRI spike on Nhava Sheva-Felixstowe lane. Best reroute via Cape of Good Hope with +18hr ETA delta.

5.4 Anticipated Judge Questions & Answers

How does DRI work?	Three models: XGBoost classifies current risk features, LSTM predicts 6-hour temporal trend, Isolation Forest detects anomalies. Ensemble fuses scores into a 0-100 index.
Is this real data?	In the MVP, data is seeded using real port coordinates and realistic disruption patterns. Full version ingests live APIs — weather, AIS vessel tracking, port authority feeds.
What makes this different from existing tools?	FourKites and project44 provide visibility. Precursa provides action. We don't just alert — we reroute autonomously before the operator even sees the notification.
How do you make money?	SaaS subscription: Starter $2K/month (500 shipments), Growth $8K/month (5,000 shipments), Enterprise custom. First target: mid-size freight forwarders.
Are the impact numbers real?	Projected based on simulation. $47K per early-detected disruption is based on industry benchmark cost models. Pilot validation is Phase 1 objective.

6. 50-Hour Build Plan

This section provides the hour-by-hour execution plan for the hackathon. The plan is structured for a backend-strong team starting from zero. Critical path items are marked.

6.1 Team Task Allocation

Role	Primary Tasks	Hours	Owner
Backend Lead	FastAPI server, data models, WebSocket, disruption simulation endpoint	Hours 0-30	Team Member 1
ML Engineer	XGBoost model training, DRI scoring logic, data generation scripts	Hours 0-20	Team Member 2
Graph Engineer	NetworkX port graph, rerouting algorithm, route ranking logic	Hours 8-28	Team Member 1/2
Frontend Dev	React dashboard, Leaflet map, DRI cards, disruption UI	Hours 20-42	Team Member 3
Integration	Claude API Copilot, end-to-end testing, demo seeding	Hours 38-48	All

6.2 Hour-by-Hour Schedule

Hours	Phase	Tasks	Priority
0 – 8	Backend Core	FastAPI project setup, shipment data models (Pydantic), mock data generator (5 shipments, realistic port coords), /api/shipments endpoint, basic WebSocket scaffold	CRITICAL
8 – 20	ML + Graph	Download Kaggle logistics disruption dataset, train XGBoost classifier, DRI scoring function, NetworkX graph with 18 port nodes (real coordinates), modified Dijkstra with cost+ETA+risk weights	CRITICAL
20 – 30	API Integration	/api/trigger-disruption endpoint, /api/reroute/{id} endpoint, WebSocket real-time push working end-to-end, DRI update loop (30-second polling)	CRITICAL
30 – 38	Frontend	React app scaffolding, Leaflet map with port markers, shipment marker layer with DRI color coding, disruption simulation panel, route visualization on map	HIGH

Hours	Phase	Tasks	Priority
38 – 44	Copilot + Polish	Claude API integration in frontend, Copilot chat panel, disruption toast notifications, route comparison sidebar, DRI trend sparklines	HIGH
44 – 48	Demo Seeding	Seed all 3 demo scenarios with realistic data, end-to-end test all demo steps, fix UI polish issues, rehearse demo script 3 times	HIGH
48 – 50	Buffer	Bug fixes, final rehearsal, prepare Q&A answers, backup demo recording in case of live issues	SAFETY

6.3 Data Sources & APIs

XGBoost Training Data	Kaggle: 'Supply Chain Shipment Pricing Dataset' or 'Freight Delay Dataset' — free, no API key
Port Coordinates	Use real coordinates for 18 major ports: Shanghai, Singapore, Rotterdam, Hamburg, Los Angeles, Mumbai, Nhava Sheva, Colombo, Jebel Ali, Hong Kong, Busan, Felixstowe, Antwerp, Tanjung Pelepas, Port Klang, Chennai, Sydney, Durban
Weather Simulation	Seed realistic weather disruption events from historical data (no live API needed for MVP)
Claude API	Anthropic API key — use claude-sonnet-4-20250514 model for Copilot responses
Leaflet.js	Free, no API key required, OpenStreetMap tiles included

7. Market Analysis

7.1 Market Size

Market	Size	Year	Source
Global Supply Chain Disruption Losses	$184 Billion/year	2024	Resilinc Industry Report
Supply Chain AI Market	$10.1 Billion	2024	MarketsandMarkets
Supply Chain AI Market (Projected)	$23.6 Billion	2028	MarketsandMarkets
Supply Chain Visibility Market	$6.8 Billion	2025	Grand View Research

7.2 Target Segments

- Global manufacturers with multi-tier supply chains — highest disruption exposure, highest willingness to pay
- Freight forwarders and 3PL logistics providers — need real-time visibility across thousands of shipments
- E-commerce giants with high SLA sensitivity — every delayed shipment impacts NPS and customer retention
- Pharma and cold-chain operators — zero-tolerance for delays, regulatory compliance requirements add urgency

7.3 Competitive Landscape

Competitor	What They Do	Limitation	Precursa Advantage
FourKites	Real-time shipment tracking and visibility	Visibility only — no autonomous action	Precursa detects AND fixes autonomously
project44	Supply chain visibility platform	Alert-based, requires manual response	Precursa reroutes before operator is alerted
Resilinc	Supply chain risk mapping	Strategic risk only, no real-time ops	Precursa operates at shipment level in real time
Flexport	Digital freight forwarding	Carrier-specific, not network-wide	Precursa is carrier-agnostic, network-wide

7.4 Projected Impact Metrics

The following metrics are projected based on industry benchmark cost models and pilot simulation results. These will be validated through live pilot partnerships in Phase 1.

- $47,000 saved per early-detected transoceanic disruption (based on industry average cost-per-disruption models)
- 40% reduction in expediting costs through automated rerouting vs. manual emergency rebooking
- 3x faster incident response compared to manual operations teams
- 60% fewer customer SLA breaches in pilot simulations

8. Business Model

8.1 Revenue Model — SaaS Subscription

Tier	Monthly Price	Shipment Limit	Key Features	Target Customer
Starter	$2,000 / month	Up to 500	Dashboard, DRI alerts, manual rerouting	Mid-size freight forwarders
Growth	$8,000 / month	Up to 5,000	Autonomous rerouting, Copilot, API access	Regional 3PLs, e-commerce
Enterprise	Custom pricing	Unlimited	Custom integrations, SLA, dedicated support	Global manufacturers, pharma

8.2 Unit Economics

Target CAC	$12,000 – $18,000 (enterprise outbound sales + pilot program)
Target LTV	$96,000+ (Growth tier, 12-month avg. contract, 85% retention)
LTV:CAC Ratio	5:1 to 8:1 at scale (healthy SaaS benchmark is 3:1+)
Gross Margin	78–82% (infrastructure costs scale sublinearly with customers)
Payback Period	4–6 months per customer at Growth tier

8.3 Go-To-Market Strategy

Phase 1: Pilot-First Sales (Months 0–6)

Target 3 mid-size freight forwarders in India and Southeast Asia. Offer a 90-day free pilot in exchange for data sharing and testimonial rights. Convert to paid Growth tier post-pilot. This validates impact metrics and builds credibility for enterprise sales.

Phase 2: Inbound + Partnerships (Months 6–12)

Content marketing on supply chain disruption topics, conference presence at LogiSym, TOC Asia, and Supply Chain India. Partner with ERP vendors (SAP, Oracle) as an embedded intelligence module.

Phase 3: Enterprise & Global (Months 12–18)

Direct enterprise sales to pharma, FMCG, and automotive manufacturers. Series A fundraise to fund global expansion into EU and US markets.

9. Product Roadmap

The roadmap is structured in three phases, progressing from MVP validation to full platform deployment to global scale. Each phase has clear deliverables, success metrics, and funding milestones.

9.1 Roadmap Overview

PHASE 1 — Validate	PHASE 2 — Build	PHASE 3 — Scale
Months 0 – 3 MVP live and pilot-validated • Kafka + Flink pipeline live • DRI scoring for 5 disruption types • Dashboard with real-time map overlay • Pilot with 1 logistics partner • Basic autonomous rerouting	***Months 3 – 9*** Full platform deployed • Full ML ensemble on SageMaker • LangGraph autonomous rerouting agent • API integrations: carriers, customs • Carbon-aware routing module • 5+ pilot customers, $50K MRR target	***Months 9 – 18*** Global expansion • Digital Twin simulation engine • Industry modules (Pharma, FMCG, Auto) • Federated learning across partner networks • Series A fundraise • EU + US market entry

9.2 Success Metrics by Phase

Phase	Technical Milestone	Business Milestone	Funding Milestone
Phase 1	MVP live, DRI accuracy >80% on test set	1 pilot partner, impact data collected	Pre-seed / hackathon prize + grants
Phase 2	Full ensemble deployed, <2 min detection confirmed in production	5 paying customers, $50K MRR	Seed round: $500K – $1M
Phase 3	Digital twin live, federated learning operational	30+ customers, $500K MRR	Series A: $5M–$10M

10. Risk Analysis

10.1 Technical Risks

Risk	Level	Impact	Mitigation
XGBoost model underperforms on real data	High	DRI scores inaccurate, rerouting suboptimal	Train on diverse dataset; add human override; use rule-based fallback DRI
Graph rerouting produces infeasible routes	Med	Demo breaks at critical moment	Pre-validate all 3 demo scenarios end-to-end; hardcode fallback route as safety net
Claude API latency too high for live demo	Low	Copilot response feels slow	Pre-cache one response; show loading state; response in 2-4 sec is acceptable
WebSocket connection instability	Med	Real-time map updates freeze	Add polling fallback; test on event WiFi network in advance

10.2 Business Risks

Risk	Level	Impact	Mitigation
FourKites / project44 copies the autonomous action layer	Med	Competitive moat eroded	Move fast on pilot data; proprietary DRI model becomes moat; file provisional patent
Enterprise sales cycle too long (12–18 months)	High	Cash burn before revenue	Lead with SMB freight forwarders (3–6 month cycles); pilot-to-paid conversion model
Carrier API integration complexity	High	Full automation layer delayed	Phase 1 uses operator-confirmed actions; automation is Phase 2 with real integrations

11. Pitch Slide Amendments

The following amendments should be made to the existing Precursa pitch deck before presentation. These changes increase credibility without reducing ambition.

11.1 Required Changes

Slide	Current Text	Change To	Reason
05	'Proven Impact' heading	'Projected Impact (Simulation-Based)'	Avoids credibility damage if judges probe validation
03	No framing of MVP vs vision	Add: 'Phase 1 MVP: Live demo. Full vision: 18-month roadmap.'	Judges respect honesty; shows you know the scope
New slide	No business model slide	Add: SaaS pricing table (Starter/Growth/Enterprise)	Critical gap — judges will ask about monetization
New slide	No competitive analysis	Add: Competitor table vs FourKites, project44, Resilinc	Judges in supply chain track will know these players

11.2 Slides to Keep As-Is

- Slide 01 — Cover: Strong visual identity, clean tagline
- Slide 02 — Problem Statement: Excellent stat framing, keep all three pain points
- Slide 04 — Tech Stack: Architecture diagram is impressive, keep as the full vision
- Slide 06 — Roadmap: Good three-phase structure, minor update to add success metrics
- Slide 07 — Closing: 'Every minute undetected costs $2,500' is a strong closing hook

12. Closing & Verdict

12.1 Is Precursa Win-Worthy?

Yes — conditionally. The idea is genuinely strong: real problem, large market, technically sophisticated vision, and a compelling autonomous action layer that competitors lack. The 'self-healing' narrative is sticky and differentiating.

The win condition is executing the demo flawlessly. A judge who sees a live shipment get disrupted, watches the DRI spike, observes autonomous rerouting, and then asks the Copilot a follow-up question — that judge will remember Precursa. The technology story becomes real in that moment.

12.2 The Three Things That Will Win It

#	What	How
1	The Live DRI Spike	Trigger the disruption simulation live. Watch a green shipment turn red in real time. This one moment shows the problem and solution simultaneously.
2	The Autonomous Reroute	The map route visibly changes without anyone clicking a button. Judges will lean forward. Let the silence land before you explain what happened.
3	The Live Copilot Question	Ask the Copilot a follow-up question during the demo. 'Why Colombo over Jebel Ali?' A real AI answer in front of judges transforms a prototype into a product.

12.3 Final Word

Every minute a supply chain disruption goes undetected costs $2,500. Precursa finds it in under 120 seconds — and fixes it. Build the demo, nail the three moments, and answer the monetization and competition questions confidently. That is the path to winning.

Predict Hours before impact	**Reroute** Graph-optimized instantly	**Heal** Autonomous & self-correcting